TORSION KERNEL AND COKERNEL DO NOT CHANGE THE RANK OF A MODULE

THE STACKS PROJECT AUTHORS

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Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- Torsion element, torsion module and the fraction field K of a domain. definition environment at source lines 4781-4792 of the same chapter and the discussion at source lines 4794-4799. Supplies the hypothesis that kernel and cokernel are torsion and the field over which the ranks are computed.
- A torsion module M has M tensor K equal to zero. used at proof lines 5034-5035, the hinted vanishing of the tensored kernel and cokernel. Kills both the kernel and the cokernel after tensoring with K, which is what makes the tensor map an isomorphism.
- Rank is the dimension over K of M tensor K. definition environment at source lines 5015-5019 of the same chapter. Converts the isomorphism of K-vector spaces obtained from the tensor sequence into the equality of ranks asserted by the lemma.
- Right exactness of the tensor product. used at proof lines 5034-5035 to obtain the exact sequence M tensor K to M' tensor K to M" tensor K to 0. Produces the K-linear map between the two ranks whose isomorphism is to be shown.

Source reference. Original lemma 0H9V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21540–21540. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a domain. If $M \to M'$ is a map of R-modules whose kernel and cokernel are torsion, then the rank of M equals the rank of M'.*

Proof. Omitted. Hint: the induced map $M \otimes_R K \to M' \otimes_R K$ is an isomorphism if K is the fraction field of R. □